

Cole Becker

📍 Cambridge, MA | ✉ colbeck@mit.edu | 🌐 colbeck | 🎓 Google Scholar | 🌐 colebecker.me

Summary

MIT PhD candidate developing theoretical foundations and practical applications of diffusion models for optimization and robotics. Research spans reward-aware sampling, multi-modal learning, and flow-based generative models. Seeking summer 2026 internship in generative AI.

EDUCATION

Massachusetts Institute of Technology

Sep. 2023 – Present

PhD Student in Computer Science – Advisor: Pablo Parrilo

Cambridge, MA

- My research focuses on two areas: 1) adapting pre-trained generative models for reward-aware sampling 2) learning across different data sources to improve sample quality in target domain.

Princeton University

Sep. 2017 – May 2023

M.Eng. and B.S.E. in Electrical and Computer Engineering

Princeton, NJ

Advisors: Bartolomeo Stellato and Amir Ali Ahmadi

- **GPA:** 3.79/4.0 - *Magna Cum Laude*
- **Thesis:** Learning Uncertainty Sets in Robust Optimization

EXPERIENCE

Diffusion Models for Optimization

Fall 2023 - Present

Massachusetts Institute of Technology

Cambridge, MA

with Professor Pablo Parrilo and Professor Asu Özdağlar

- Developed flexible diffusion-based optimization framework for (i) solving inverse problems with diffusion, (ii) sampling from tilted (reweighted) distributions, and (iii) optimizing reward functions over potentially low-dimensional distribution manifolds, without the need to train task-specific conditional models.

Diffusion for Robotics

Fall 2024 - Present

Massachusetts Institute of Technology - Professor Russ Tedrake

Cambridge, MA

- Developing flow-based sim-to-real transfer methods for diffusion policy training in robot manipulation.

Robust Optimization

2022 - 2023

Princeton University - Advised by Professor Bartolomeo Stellato

Princeton, NJ

- Co-led a project to build a robust extension package to cvxpy to automate dualization from robust optimization problems to tractable convex programs.

JOURNAL PUBLICATIONS

Mean Robust Optimization

2023

I. Wang, C. Becker, B. Van Parys, B. Stellato

- *Mathematical Programming* 213 (1), 1235-1277
- **Best Paper Award** ICCOPT 2022

Learning Decision-Focused Uncertainty Sets in Robust Optimization

2023

I. Wang, C. Becker, B. Van Parys, B. Stellato

- arXiv preprint arXiv:2305.19225

TEACHING

Instructor – Diffusion Models

Massachusetts Institute of Technology

Jan. 2025, 2026

Cambridge, MA

- Co-taught introductory January course in Diffusion Models, lectured, designed problem sets, and led office hours.

Assistant Instructor – Convex Optimization and Other Courses

Princeton University

Fall 2020 – Spring 2022

Princeton, NJ

- Served as Assistant Instructor and Graduate/Undergraduate TA for multiple courses including Convex Optimization, Convex & Conic Optimization (PhD), and Introduction to Logic Design. Responsibilities included leading weekly office/study hours, grading and writing assignments and exams, delivering review lectures

FELLOWSHIPS AND AWARDS

National Science Foundation Graduate Research Fellowship (NSF GRFP)

2023

George Mueller Award – Princeton University

2022

Highest achievement across engineering and athletics

TALKS

ICCOPT

Learning Uncertainty Sets in Robust Optimization

July 2022

Lehigh University

SKILLS

Software : Python, MATLAB, CVXPY, PyTorch, JAX, NumPy, Pandas, Seaborn, Matplotlib, Java

Languages : English (native), French (B1), Spanish (B1)

ACTIVITIES

Squash

2008 – Present

- Princeton Varsity Squash Team – Captain (2020–21)
- Professional Squash Player (Top 5 U.S.)
- U.S. National Squash Team Member (2015–2021)